Team 51

Nerd Herd:
INSOMNIACS



Introduction

Our Team:

Team 51 CAMS Nerd Herd Insomniacs is a FIRST Tech Challenge (FTC) team based in Carson, California.



Table of Contents:

Introduction 1
Meet Our Team 2
Team Plan 3
Community Outreach 4-5
Design Principles 6
Iterative Intake. 7
Robot Iteration. 8
Design 9
Robot Overview. 10-11
Manufacturing 12
Game Strategy 13
Programming 14
Experiences and Learning 15

Our Goal:

At our school, Nerd Herd strives to provide students with hands-on, real-world experiences in engineering and business. Outside of the classroom, Team 51 aims to spread STEM values throughout the community. We hope to inspire the next generation of problem solvers, go-getters and creative thinkers. Specifically within our team, we strive to create an environment that emphasizes teamwork, respect, innovation, commitment and gracious professionalism.

Our team focuses on developing the technical and personal skills of our members. We focus on design principles and processes, moving from concept to finished product. In our community, we consistently demonstrate our core FIRST values and gracious professionalism when working with others. We strive to increase the impact of FIRST and STEM through awareness. Additionally, we strengthen connections with technical experts and other members of the STEM community through collaboration.

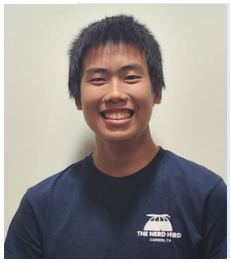
Meet Our Team!

FTC Team 51 Insomniacs is composed of 8 students, 1 parent coordinator, 1 parent mentor, and 1 team lead. The whole team branches out to perfect each of our 6 subteams (see page 4, learning section for subteams).



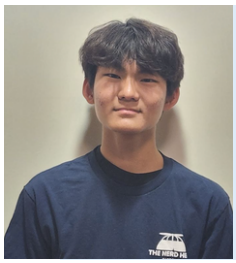
Sarah Arata

Team Lead



Felix Li

Assembly
Design



Kenneth Lee

Driver
Admin
Assembly



Gabriel Duenas

Assembly,
Driver
Design



Aanya Tripathi

Admin
Assembly
Outreach



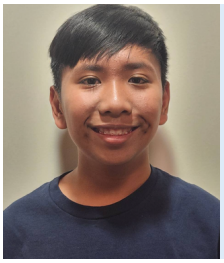
Sara Sahdev

Assembly
Outreach



Zachary Martinez

Programmer



Anthony Wheeler

Programmer
Outreach



Bhargav Ayyagari

Assembly



Jaspal Sahdev

Parent
Coordinator



Doug Martinez

Parent
Mentor

Team Plan

Overview

Our goals that we hope to achieve going into this year's FTC season, are to gain more experience with the field of STEM, build upon communication skills across the team, and to solve problems with a critical mindset.

Our sponsors: Boeing, Raytheon, Raising Canes, Honda

Learning

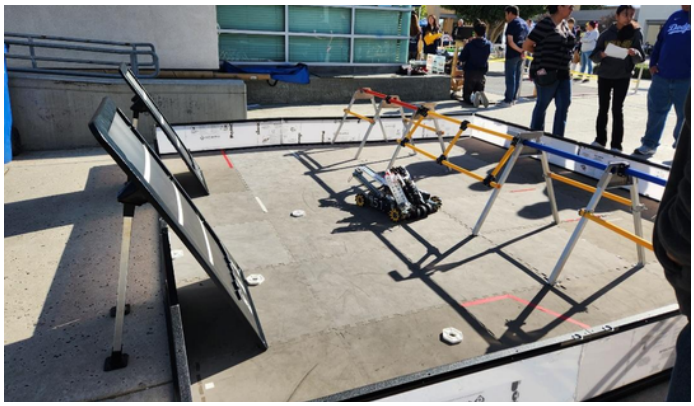
At first, our team was experiencing difficulties in team organization and task efficiency, which slowed us down. However, we began to communicate as a group, allowing us to explore the skills of each of our members and sort them into positions that would capitalize on those skills, creating our 6 subteams: Programming, Admin, Assembly, Outreach, Design, and Drive. This allowed everyone to have a clear view of their responsibilities and allowed us to efficiently complete tasks and reach our goals.

Community Outreach

Taste Of CAMS:

- Taste of CAMS which is an annual school-wide event where we fundraise and showcase our clubs, interest groups, and teams.
- We were able to demostrate our robot and team to over 120 prospective students and parents seeking to apply to CAMS





CAMS Site Day:

- During this event, we showed off our team and robot to over 80 middle school students.
- We allowed them to take turns operating our robot under supervision to build hands-on experience and interest in robotics.

Instagram:

- We created an Instagram account to reach out to our community over the internet
- We made posts displaying our progress, our team, our competitions, and the fun moments we had along the way to inspire others to join FIRST.
- Follow us at @ftc_51 !



Contact Information of the Teachers of the classes in which we visited:

- Mr. Christian Herrera: Room #201, 3rd Grade, cah1898@lausd.net
- Ms. Urzula Nelson: Room #201, uln6469@lausd.net
- Ms. LaFortune: Room #110, 1st Grade, axl2367@lausd.net
- Ms. Monica Weston: Room #4, 5th Grade, monica.weston@lausd.net
- Mrs. Mossberg: Room 4, 1st Grade, mbm5861@lausd.net

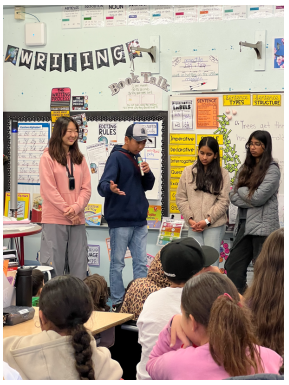
Community Outreach

Impact on Community

- We went to Jefferson Middle School in Torrance and South Shore Elementary in San Pedro to spread how FIRST helped us to learn/grow, along with how our robot evolved over time.
- After we finished our presentation, many kids were asking questions about what could they do to get involved. To answer this question, we promoted FLL to the teachers in the room as well as the kids.
- Impacted around 150 elementary kids.
- Ran a fundraiser for cancer awareness, raised around $500
- Ran a fundraiser at Krispy Kreme and donated profits to schools who cannot afford STEM in their schools. And we raised __ for those students in need.







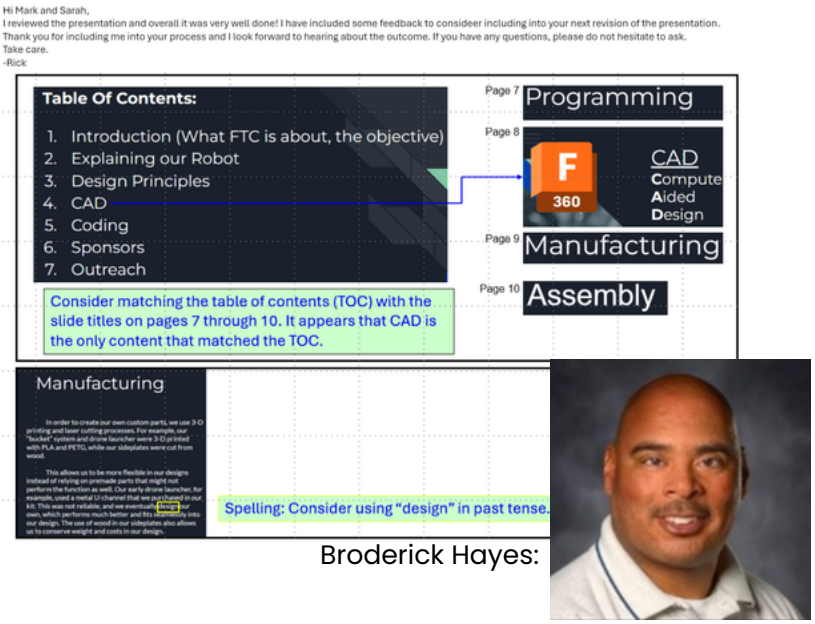

Hi Mark and Sarah,
I reviewed the presentation and overall it was very well done! I have included some feedback to consideer including into your next revision of the presentation.
Thank you for including me into your process and I look forward to hearing about the outcome. If you have any questions, please do not hesitate to ask.
Take care.
-Rick

Table Of Contents:
1. Introduction (What FTC is about, the objective)
2. Explaining our Robot
3. Design Principles
4. CAD
5. Coding
6. Sponsors
7. Outreach

Page 7 Programming
Page 8 CAD Compute Aided Design
Page 9 Manufacturing
Page 10 Assembly

Consider matching the table of contents (TOC) with the slide titles on pages 7 through 10. It appears that CAD is the only content that matched the TOC.

Manufacturing

Spelling: Consider using "design" in past tense.

Broderick Hayes:

Professionals in the Field

- We contacted industry professional Broderick Hayes, a Technical Lead Engineer at Boeing to review a presentation we gave on our team and our robot.
  - He left us with great advice on how we could improve, shown in the photo to the left.
- Jaspal Sahdev, our parent coordinator, is a computer engineer of 25+ years who helps develop the autonomous code for our robot.

CAMS Community Day

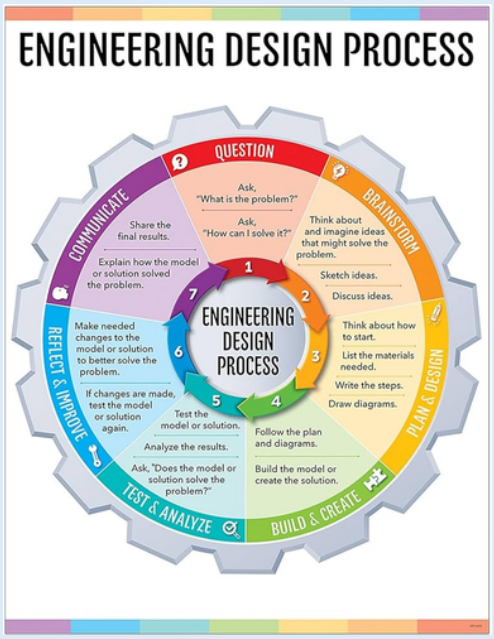
- At CAMS Community Day, our team showed our robot to the future generation of California Academy of Mathematics and Science students.
- Showing the students an inside look on how our team operates, we talked about our team dynamic and how FIRST and FTC have brought us together in and outside of robotics.
- In total, we were able reach over 50 families and invite potential students to join CAMS and FTC.



Design Principles

Here is the process we used to improve upon our intake:

Question: What can we do to improve our Intake?
Brainstorm: We could use custom flaps to intake the pixels.
Plan and Design: We use CAD to design these flaps based on the dimensions of the pixels
Build and Create: We use 3D printers to 3D print out our designs.
Test and Analyze: We apply these belts onto our skeleton for the intake, and see how efficient it is when taking a pixel.
Reflect and Improve: We realized that sometimes a pixel got stuck in our ramp when intaking, so we decided to make our belts tighter to have more force in dragging the pixels up to our ramp.
Communicate: We talked about what happened with our team.
Question: What can we do to improve our intake?



Connecting with others:

FRC:
CAMS is home to Team #687 Nerd Herd of the First Robotics Competition. Made up of upperclassmen with more experience in the field, they helped guide our thinking, manufacturing, and designing, alongside with helping us get more accustomed to the field of STEM in general.

Engineering Professionals:
We connected with the engineering teacher at our school as well as industry professionals to ask for feedback on our designs and presentation, as well as helping us secure parts and giving us tips to improve our strategy.

Mentors:
Our parent mentor Doug Martinez, helped teach our programming team how to code for the autonomous stage and gave us a tips on ways we could maximize scoring. Our parent coordinator Jaspal Sahdev, helped guide our thinking.



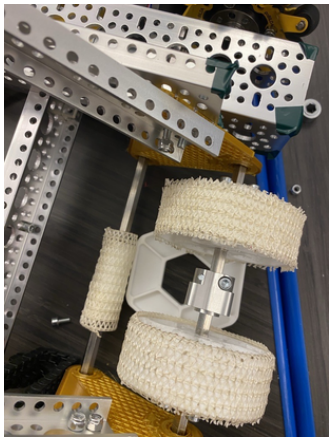
Teacher Mentor - Mr. Harder

FRC team 687

Iterative Intake:

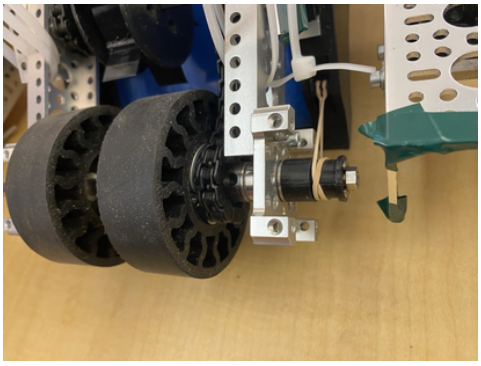
Iteration 1:

The first version of our intake used TPU wheels wrapped in anti-slip tape. TPU is a flexible material, meaning that more friction could be applied to the pixels by pushing them into the ground. The anti-slip tape also helped the intake apply friction to the pixel to pull it in. We had 2 sets of wheels, 2 large wheels that pulled pixels in, and a smaller wheel that pulled pixels off the ground. Despite this, it was difficult to get pixels off the ground, as the large wheels were fixed in place, preventing pixels from being pulled upwards.



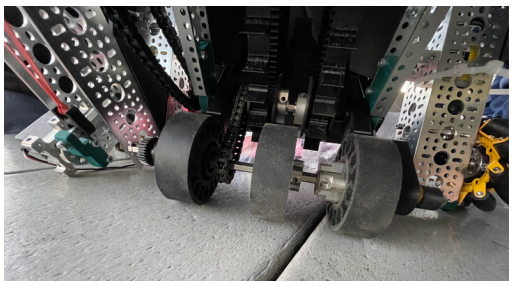
Iteration 2:

For the second version of our intake, we switched to rubber wheels rather than TPU wheels. Rubber is much more flexible than TPU, but has less friction than the anti-slip tape. Also, rather than using a smaller wheel to lift pixels off the ground, we installed a ramp. Due to the increased flexibility and the new ramp, it was now possible for the intake to lift pixels from the ground and into the bucket. Unfortunately, we realized that our intake was very thin, making it hard for the drivers to intake pixels from the wings.



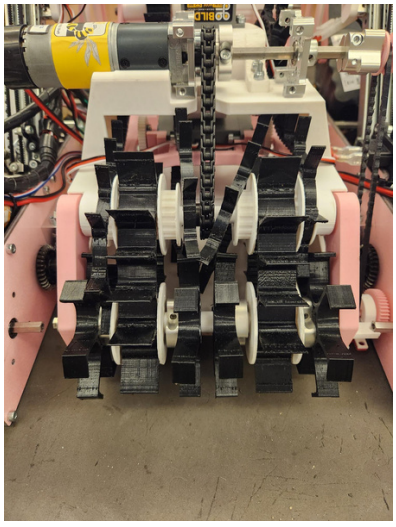
Iteration 3:

In order to make it easier for the drivers to intake pixels, for our third iteration we expanded the intake's width. We expanded the ramp and added a third rubber wheel. This meant that the driver could be much less accurate when attempting to intake pixels, decreasing cycle times and allowing us to score more efficiently. The rubber wheels had less frcition, and because of this it was hard for the drivers to get the pixels into the bucket in order to score.



Iteration 4:

Our fourth intake went back to TPU wheels, now using a "boot"-like design to "kick" pixels into the intake. We also added a guide that ensures pixels are always placed into the bucket. The bucket is also now able to hold two pixels next to one another. This meant we had to include a new mechanism to control which side the pixels were placed into. This was accomplished by blocking one side of the intake using a servo. This iteration is what we are planning to compete with at ILT.



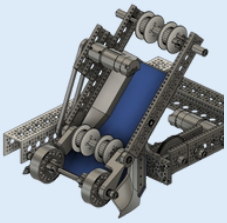
Robot Iteration

Version 1:



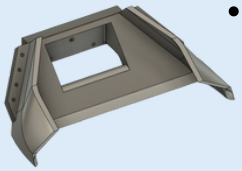
(Named: Reeree IV)

Intake CAD Model



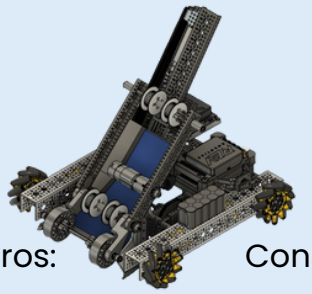
- 1 motor
- Two belts
- The motor is chained to the middle roller

Pixel Herder



- This guided the pixels into the main ramp to bring the pixels into the bucket

Full CAD Model of Robot



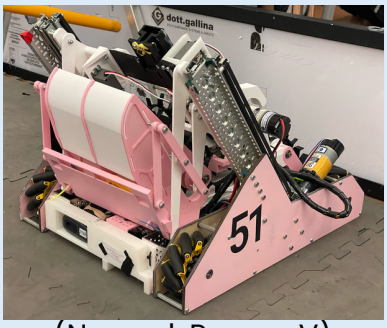
Pros:

- The robot could intake pixels semi-efficiently
- We were able to put pixels on the backdrop
- The robot could fit under the side trusses

Cons:

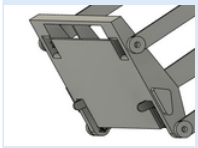
- When intaking, the pixel would bounce in the front for ~3 sec.
- The pixel wouldn't load into the bucket accurately
- The extending arm would bounce, proving inaccurate
- Not able to fit under door

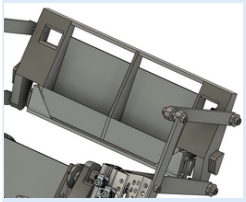
Version 2:



(Named: Reeree V)

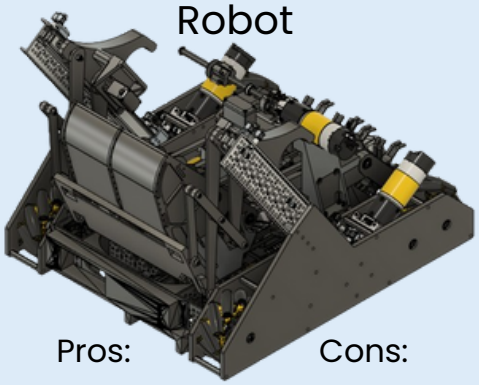
Passive Outtake Mechanism





- Passive
- Press against backdrop to release pixels
- Rubber band powered
- Keeps pixels in without mechanism

Full CAD Model of Robot



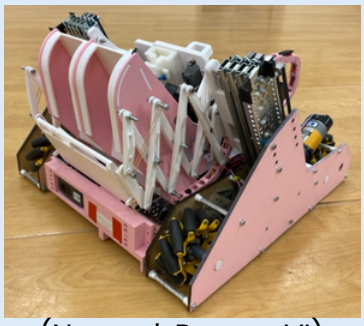
Pros:

- The lift was very stable
- The intake worked mostly reliably
- We were able to reliably hang
- Passive outtake
- Odometry wheels
- The robot could fit under the side trusses
- Able to pick up from the stacks

Cons:

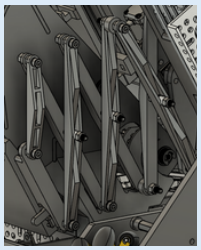
- Not able to direct which track the pixel went down in the intake
- Heavier
- When intaking, the pixel would bounce in the front for ~3 sec.
  - This problem persisted
- Not able to fit under door

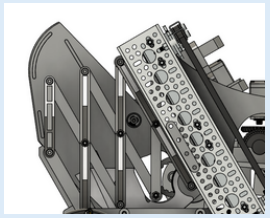
Version 3:



(Named: Reeree VI)

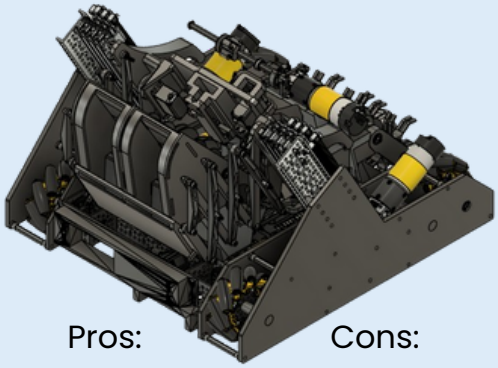
Scissor Lift





- 2 servos
- 10-bar scissor lift
- Angled based off linear slides
  - Linear slides are parallel with the backdrop, which makes the end of the scissor lift flush with the backdrop

Full CAD Model of Robot



Pros:

- Stable Lift
- Drone Launcher is mainly consistent in getting zone 1
- Reliably hang
- Can fit under door
- Can outtake on backdrop
- Faster
  - We added new motors
- Able to direct which track the pixel went down in the intake

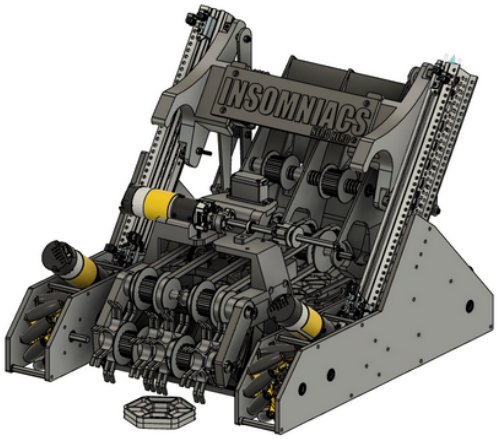
Cons:

- Still heavy
- Can't control how many pixels are picked up from the stacks

Design

Our Robot Design

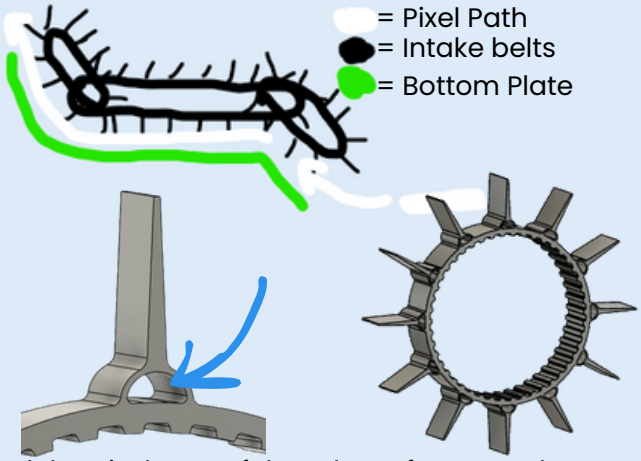
- In the process of creating our robot, the use of CAD (computer aided design) has been the backbone of our efforts.
- CAD has let us design each and every mechanism, component, and system that is used on the robot.
- We were able to format our ideas into a model, then use the model to bring our ideas into a reality
- A prominent example of intensive design work would be our intake



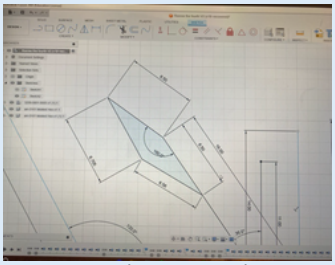
Second Iteration Robot (Named Reeree V):

Design Quirks

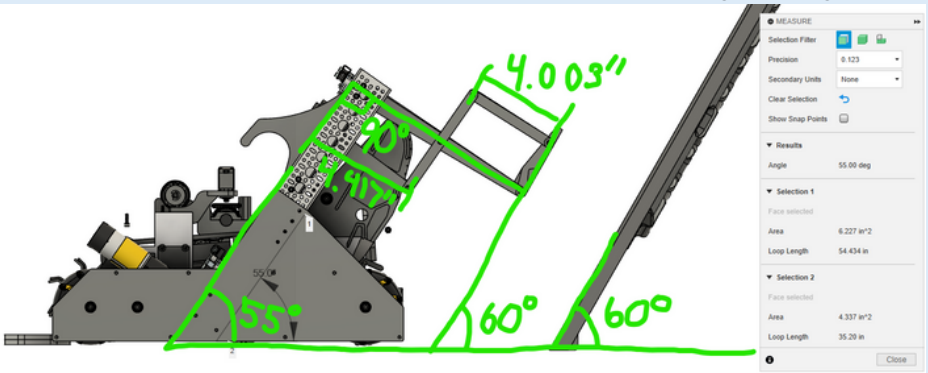
- Our Intake brings the pixel up and into the robot by using 3D printed belts with flaps.
  - We made measurements to calculate exactly how long the belts needed to be to fit tightly with the pulleys, as well as the sizing of the individual teeth of the belt for it to fit snugly with the teeth of the pulley
  - At the base of each belt there is an empty area. This is here because when the belt curves around the pulley, the two thin connection points are stretched away from each other, resulting in the final belt being able to curve more easily compared to if it was all filled in.
  - The belts are 3D printed with a flexible filament called TPU



- Our lift makes use of a modified scissor lift mechanism to extend the pixels out of the robot's frame and onto the backdrop (This applies to Version 2. Version 3's lift is able to be parallel with the backdrop)
  - Our self imposed sizing restraint in the vertical domain is 14".
  - We knew we needed to use two linear slides, but we couldn't attach them in a way that would be parallel with the backdrop without straying past the 14" restriction because we can't have them scraping the ground, and we needed to fit our odometry wheels into our robot's side plates.
  - We ended up needing to put them at a 55 degree angle instead of the backdrop's 60 degree angle.
    - Though this may seem like a harmless choice, it was far from such because if we then used a regular mechanism that kept the end attachment parallel with the lift, we would be 5 degrees off from the board, which could've let the pixel not be scored or bounce off the bottom.
  - We implemented a scissor lift with one beam shorter than the rest
    - To do this, we used calculations to make it so that when the scissor lift is 90 degrees from the linear slides, the bucket (which drops the pixels downwards) would be at a 60 degree angle from the ground.

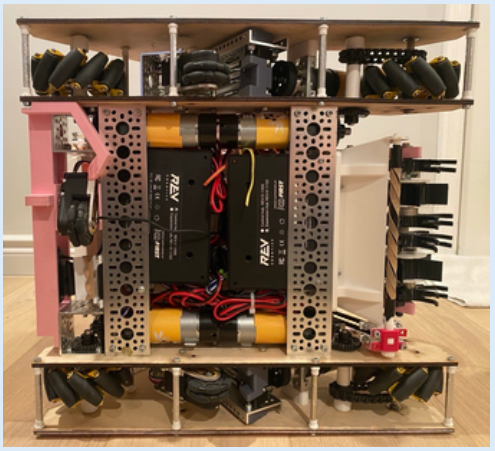


Early Size Testing:



Robot Overview

Drive Train:



- 4 mecanum wheels
  - Allows for higher levels of maneuverability
- 435 rpm GoBilda Yellow Jacket motors
  - We originally had 312 rpm motors until we decided our pixel cycle time would be increased with faster motors
- Robot centric drive achieved using odometry wheels
  - The drive code is based on the robot's right and left
- Wooden side plates
  - Wood is lighter and easier to work with in comparison to aluminum

Intake:

- The intake is driven by one 315 rpm GoBilda Yellow Jacket motor
  - The whole system is belted, chained, and geared together
- The intake is comprised of almost entirely custom designed 3D printed parts
  - The front rollers, along with the custom sized belts and the custom belts with flaps are all printed using TPU, a flexible/bendable 3D printing filament (The black belts are printed with TPU)
  - The front ramp, along with the middle flat ramp and the pixel orientator are all 3D printed using PETG, a stronger and more resilient alternative to PLA filament. (Everything that is white is 3D Printed in PETG)
  - The two beams that hold the front rollers in place, as well as the plates that hold the pixel orientator upright and in the correct position is 3D printed using PLA.(Everything that is pink is 3D Printed in PLA)
- The intake makes use of a bottom roller to prop up the pixels when they are being intaken (Version 1 and 2's intakes weren't efficient, so we decided to try something new here)
  - The roller is on a 3:1 ratio from the driving motor, therefor going faster than the top TPU rollers
  - It is wrapped in 3 cut rubber bands to help heighten friction between the pixel and the roller
  - A custom sized TPU belt used to drive the bottom roller; Now the roller is being driven by a chain
- Once the intake has taken in the pixel, it needs to be diverted into one of two tracks (This would allow us to make mosaics with higher accuracy)
  - We use pixel diverters powered by two servos to control which side of our intake the pixel gets herded into
  - They make use of a rack and pinion system
- After traveling down a track, the pixel must make its way into our passive outtake mechanism by using our orientator
  - After extensive brainstorming, we have come up with a design that would let the pixel fall into our 60 degree angled outtake mechanism
  - The pixel would travel up the end ramp, and into the orientator, then fall back downwards into the open bucket (outtake mechanism)
- We had planned on implementing passive C-shaped mechanisms that would act as a heart valve to prevent two pixels from going down the same track, but there was not enough time to implement these designs



Front TPU Rollers + PLA & PETG beams



Intake track + Belts with flaps



Pixel Orientators

Passive Outtake Mechanism:

- The passive outtake mechanism has two parts
  - Main Pixel Housing
    - This is where the pixels drop into after the intake has transferred them across the robot
    - This is also where the scissor lift (coming up next) will connect to
  - Main Housing's Floor
    - This small part slides into the main housing, being held in place with two rubber bands
    - The floor piece has two prongs which, when the driver pushes the bucket into the backdrop, forces the floor to retreat backwards, therefor dropping the pixels that were in the main housing further down the backdrop
- We chose a horizontal outtake mechanism in comparison to a vertical one because we believe that it would be more efficient in cycle time when scoring pixels compared to having to take more time to aim where the pixel would be released in a vertical outtake mechanism



Two pixels in the main housing area



The side that contacts the backdrop

Scissor Lift:

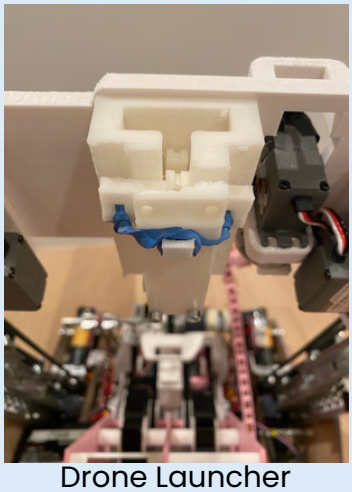
- Two servos power an angled scissor lift
  - The scissor lift is connected to the passive outtake mechanism, lifting it past the confines of the robot's perimeter and bringing it higher than the linear slides can reach
- Two rubber bands help the scissor lift fall back into the correct loading position
  - Originally, there was some leeway in the scissor lift's downwards position, which left an inch wide gap in between the intake orientators and the passive mechanism
  - Strategically placed rubber bands ensure the correct loading position of the Passive Outtake Mechanism in accordance to the Intake
- The scissor lift's individual beams are designed to be low profile while still retaining strength in their supposed "flat" axis
  - The beams have ridges, acting like supports for a bridge



Scissor lift extended

Drone Launcher:

- One servo actuates a rubber band powered drone launcher
  - The drone launcher uses a rail system to guide the backing of the paper airplane forwards at rapid speeds
- The mounting plate of the drone launcher is attached to the servo mounts of the scissor lift, effectively holding the two linear slides together while staying clear of the hanging hooks
- The mounting plate is offset forwards because it needed to fit in between the belts of the intake while also not running into the intake dividers
- The compact size of the drone launcher lets us use the extra space for the intake



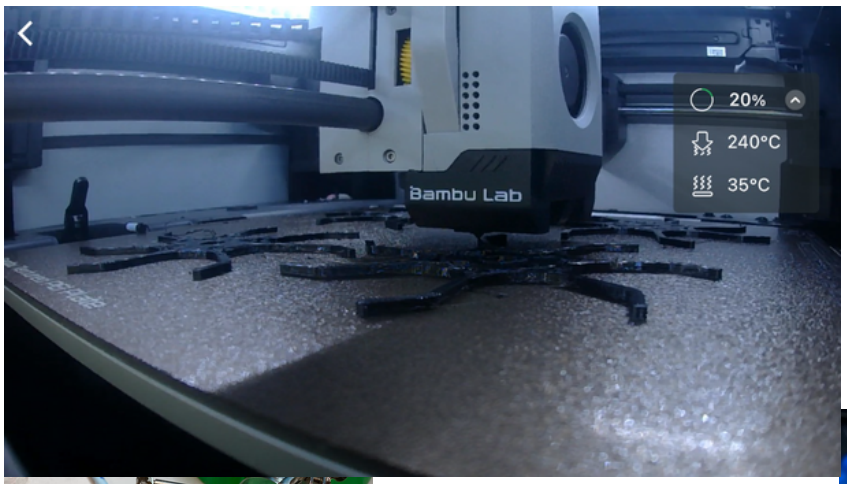
Drone Launcher

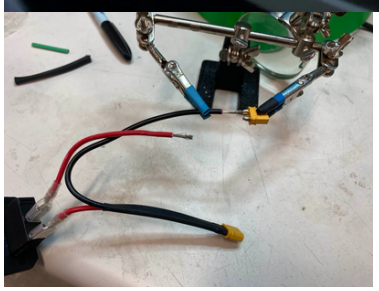
Manufacturing

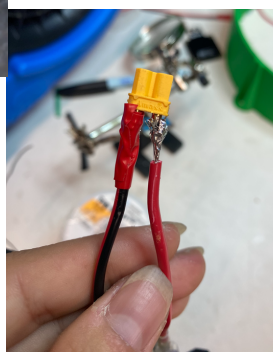
3D Printers: We utilized several 3d-printed parts in our robot's design. We needed many custom parts, and 3d printing was the most affordable and most practical option for us. This let us print the majority of our robot's mechanisms, such as the TPU wheels and belt in our intake and the PETG hooks to let us hang onto the truss.

Laser Cutter: We used a laser cutter to manufacture our wooden side plates. Laser cutting is highly accurate compared to hand cutting. In previous iterations of our robot, we laser cut polycarbonate for our intake's ramp. Polycarbonate is a type of low friction plastic, meaning pixels could slide on it with little resistance.

Soldering: In some cases, we had to solder wires together to ensure a consistant connection, especially on moving or damaged parts.



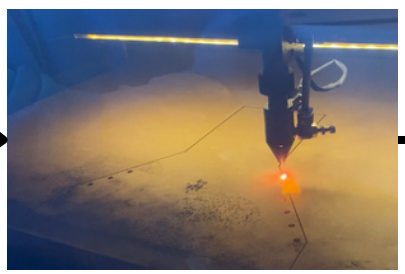








CAD model of Ramp:



Laser cutting ramp out of polycarb:



Final product:

Game Strategy

Autonomous:

- Purple Pixel placed next to Team Prop (20 points)
- Yellow Pixel placed on backdrop according to Team Prop (25 points)
- Parking (5 points)
- 50 Points

2 Minute Driving Control:

- We focus on forming mosaics, three pixel groups of either the same color or three different colors. These mosaics score bonus points with the artist bonus.
- At the start of the driver control period, we attempt to create a mosaic with the yellow pixels already placed on the backdrop during the autonomous period.

- Aiming for at least 2 mosaics (20)
- 1 Setline (10)
- 12 Pixels (36)
- Averaging Around 60 Points

Last 30 seconds: End Game

- Launch drone (Depends on landing zone, up to 30 points)
- Hang on the trusses (20 points)
- 20 to 50 Points

Alliance Strategies

During our meets and our alliances, we changed our strategy and our way of working together with our allaince based on the strengths of our teams. For example, when we teamed up with Team 5390 during Meet 3, we decided that they would be placed closer to the backboard due to them having a better autonomus program, and because they usually need more time to score on the backboard. With our improved robot, we decided to just maximize efficiency by making a steady cycle of scoring, while prioritizing making 2 mosaics first, then stacking white pixels to get the first set line.

Programming

Currently, we use a robot oriented mecanum drive to control our robot during Tele-Op. While driver oriented driving could be easier to control, we felt that the Control Hub's built in IMU and our motors' drive encoders were inconsistent and outweighed their benefits.

Autonomous:

We use the **Roadrunner library** written by ACME Robotics to create complex spline paths for our robot to follow during the Autonomous period. For randomization, we utilize the **ftc-ml tool and TensorFlow Object Detection** to determine the placement of the team prop. The team prop we use was specially designed by the Programming team to utilize Tensorflow's ability to recognize patterns. We then follow Roadrunner paths using a **three dead wheel odometry setup**. This three wheel setup, while requiring more parts, is much more accurate than two wheel localization with the IMU.



Our programmers are Anthony Wheeler and Zachary Martinez



Our team props, "Among Us Shrine", designed to have a distinct pattern for TFOD

Experiences & Learning

Events	What went well	What did not go well	Next Steps/What we learned
Meet 0	The team gained experience in competing in FTC competitions, and we were able to score due to intake.	Our arm was unstable and not reliable. Since Meet 0 was our first competition, our team was still inexperienced and were learning.	• Develop a more reliable way of scoring • Avoid blocking while driving • Communicate more
Meet 1	We formed our first official drive team and trained them to avoid penalties.	Our intake was inefficient and had issues picking up pixels. We also mixed up our controllers when driving.	• Improve auton/scoring ability • Devlop airplane launcher • Learned to organize our stuff better when they look similar (Maybe with labels)
Meet 2	Our drivers were well trained and our new intake efficiently picked up pixels.	Our drone launcher was inaccurate, causing drones to fly too far from the landing zones.	• Improve drone launcher • It is important to manage time
Meet 3	We were able to score many points during the tele-op period. Our intake was efficient and our drivers were quick.	Our autonomous programs were lacking. We could park in the backstage area, but couldn't complete the randomization task.	• Improve autonomus code • Make drone launcher more consistant
ILT	Auton period was able to complete the randomization task consistently. Stratergy went well and we were able to communicate with our alliences beforehand.	Many of our members were blind sighted by the judge's questions and had trouble answering them. Also, we realized our driving was still relatively slow.	• Practice interviews and review key information. • Focus on speed during tele-op